

WHICH CEREALS ARE THE BEST RATED? AND WHY?

Pt. 1

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Project Purpose

Project Question: What nutritional factors most accurately predict a high consumer rating for breakfast cereals?

This project uses course data to answer a real-world analytical question. More specifically, it investigates which factors are best for predicting a high consumer rating for breakfast cereals.

Looking at these factors through XL Miner, we can predict consumer sentiment based on a cereal's nutritional profile.

Data Description:

<https://docs.google.com/spreadsheets/d/1qkNZkhHWGB3hxxTszxukmVTeb-kVNnOVHb9E401E0Q/edit?gid=0#gid=0>

Data Cleaning, Exploration & Descriptive Analysis

a. DATA TYPES:

Nominal: Name, mfr (manufacturer), type.

Ordinal: shelf (1, 2, or 3).

Quantitative: calories, protein, fat, sodium, fiber, carbs, sugars, potassium, vitamins, weight, cups, rating

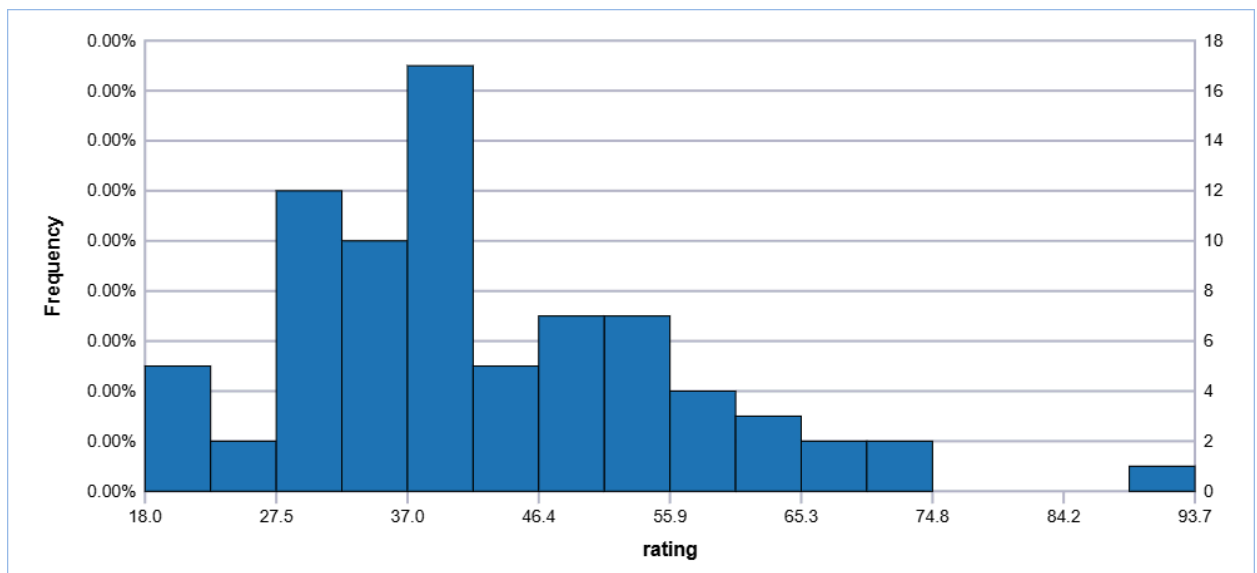
b. DESCRIPTIVE STATISTICS

sodium		fiber		rating	
Mean	159.6753	Mean	2.151948	Mean	42.6657
Standard E	9.553577	Standard E	0.27161	Standard E	1.600837
Median	180	Median	2	Median	40.40021
Mode	0	Mode	0	Mode	93.70491
Standard D	83.8323	Standard D	2.383364	Standard D	14.04729
Sample Va	7027.854	Sample Va	5.680424	Sample Va	197.3263
Kurtosis	-0.345243	Kurtosis	8.647492	Kurtosis	1.318747
Skewness	-0.575711	Skewness	2.431675	Skewness	0.91024
Range	320	Range	14	Range	75.66206
Minimum	0	Minimum	0	Minimum	18.04285
Maximum	320	Maximum	14	Maximum	93.70491
Sum	12295	Sum	165.7	Sum	3285.259
Count	77	Count	77	Count	77
Largest(1)	320	Largest(1)	14	Largest(1)	93.70491
Smallest(1)	0	Smallest(1)	0	Smallest(1)	18.04285
Confidence	19.0276	Confidence	0.540957	Confidence	3.188343

Full Descriptive Comparison:

https://docs.google.com/spreadsheets/d/1p_f2Zi-IBDfYy2PsdW_9dW1q-AnYbEbK0kZv_vp_mI/edit?gid=0#gid=0

The full descriptive statistics for all columns are very difficult to fit here. That said, we don't need it. This is because although a few things stand out and are pointed out in the image above. For one, a high standard deviation for sodium (83.83) suggests a wide variety in salt content. Second, the rating mean (42.67) is slightly higher than the median (40.40), alluding to a right skew.



c.

This histogram takes a look at the ratings just to see how they're distributed. Through this, we can see that things are fairly normalized; however, there are a good number of outliers that have the best ratings. This makes things simple. Now, my job is to figure out exactly what about them gives such high ratings.

- d. First, ratings by manufacturer. Through this, we can see which manufacturers are doing well and eventually, why they stand out among the rest.

Manufacture	Average of rating
A	54.850917
G	34.48585168
K	44.03846235
N	67.96856717
P	41.70574411
Q	42.91598988
R	41.54299713
Grand Total	42.66570499

A = American Home Food Products; G = General Mills; K = Kelloggs;
N = Nabisco; P = Post; Q = Quaker Oats; R = Ralston Purina

Highest Average Rating: Nabisco with an average of 67.97.

Lowest Average Rating: General Mills with an average of 34.49

Nabisco focuses on shredded wheat and bran, which could explain their rank as number one, while General Mills has a higher sugar content, which could explain its place at the bottom.

- e. Now, we take a look at the calorie and sugar content. This is the most obvious next step and what

Column Labels	50	70	80	90	100	110	120	130	140	150	160	Grand Total
Average of rating	72.48888967	63.914239	68.235885	60.57839629	47.43819518	35.78980803	32.8489726	33.7447025	35.25220567	35.638314	30.313351	42.66570499

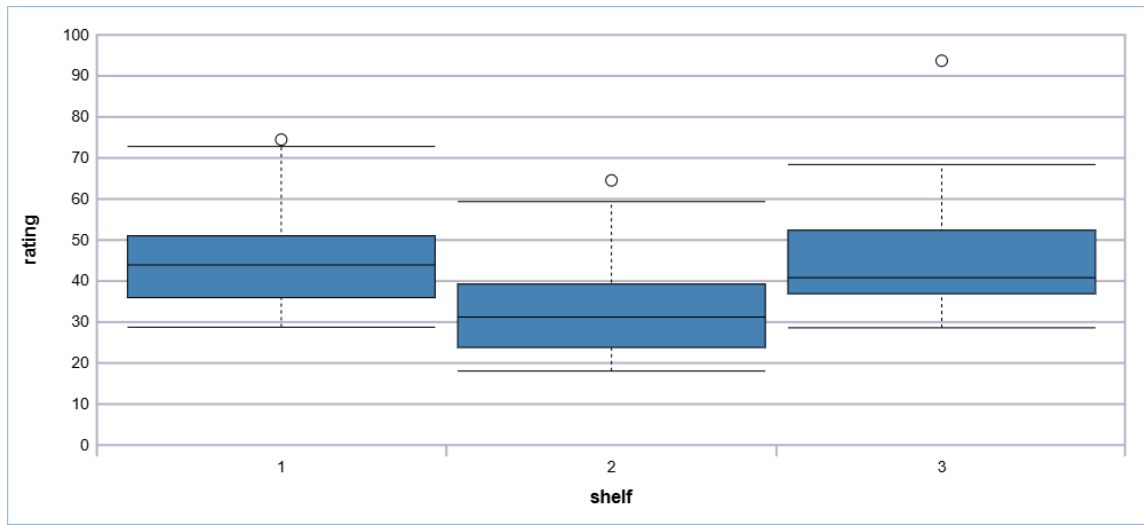
Highest Average Rating: The 50-calorie group (72.49).

Lowest Average Rating: The 160-calorie group (30.31).

We can see that as calorie content increases, the consumer rating drops. So, we see an inverse relationship, and as seen below, the manufacturers with lower sugar content usually perform the highest (data sorted by rating highest to lowest).

mfr	sugars	rating
K	0	93.70491
N	0	74.47295
N	0	72.80179
N	6	68.40297
N	0	68.23589
N	0	64.53382
Q	0	63.00565
Q	0	60.75611
K	2	59.64284

- a. Next, we take a look at the average rating per shelf to see if there's anything significant to be found.



Shelf One: You can see the highest median and an overall wide variation in distribution.

Shelf Two: This has the lowest median ratings and spread. This is most likely where the sugary cereals are.

Shelf Three: A decent variation with a very noticeable outlier. The cereals with the lower calories and sugar content are placed on this shelf (Nabisco and Kellogg). Thus, they have the higher ratings.

In other words, Shelf 3 has healthier options, and it has cereals with overwhelmingly higher ratings. There's a clear trend.

mfr	shelf	rating
K	3	93.70491
N	3	74.47295
N	3	72.80179
N	3	68.40297
N	3	68.23589
Q	3	60.75611
K	3	59.42551
A	3	54.85092
Q	3	50.82839
R	3	49.78745
Q	3	49.51187
R	3	49.12025
P	3	45.81172
Q	3	45.32807
R	3	41.44502
K	3	41.01549
P	3	40.91705
G	3	40.40021
G	3	40.10597
G	3	38.83975
P	3	37.84059
R	3	37.13686
K	3	36.47151
G	3	36.18756
G	3	36.1762
K	3	35.78279
P	3	35.25244
R	3	34.38484
R	3	34.13977

Note: Based on these experiments, we can see a great deal of evidence that healthier cereals (low sugar and low calories) have a higher rating. However, we can really put things to the test by running a linear regression on the data, highlighting what truly does and does not affect consumer rating.

Regression Analysis

Note: Through some inspiration from my instructor, I have decided to cut down on variables to calories, protein, fats, fibers, sugars, sodium, carbo, and of course, rating. I did this for two reasons: 1) All the variables listed in the original dataset, while informative, were not all necessary and were partly explained in the descriptive analysis section; and 2) These are the factors that come to mind when the general populace are thinking about purchasing cereal and food as a whole. Thus, I believe this would be the best in answering the question of "What nutritional factors most accurately predict a high consumer rating for breakfast cereals?"

a. Running the regression, I get these results:

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.995505							
R Square	0.99103							
Adjusted R	0.990106							
Standard E	1.403371							
Observatio	76							
ANOVA								
	df	SS	MS	F	Significance F			
Regressior	7	14795.37	2113.625	1073.206	0			
Residual	68	133.9225	1.969449					
Total	75	14929.29						
	Coefficient	Standard E	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
Intercept	57.79548	1.213734	47.6179	5.9E-54	55.37351	60.21745	55.37351	60.21745
calories	-0.1917	0.033546	-5.71462	2.66E-07	-0.25864	-0.12476	-0.25864	-0.12476
protein	2.72101	0.234847	11.5863	9.45E-18	2.252379	3.18964	2.252379	3.18964
fat	-2.22463	0.344971	-6.44876	1.38E-08	-2.91301	-1.53625	-2.91301	-1.53625
sodium	-0.05725	0.00216	-26.5117	1.42E-37	-0.06156	-0.05294	-0.06156	-0.05294
fiber	2.476675	0.093332	26.53605	1.34E-37	2.290433	2.662917	2.290433	2.662917
carbo	0.803899	0.152609	5.267691	1.53E-06	0.499372	1.108426	0.499372	1.108426
sugars	-1.03795	0.142229	-7.29779	4.11E-10	-1.32177	-0.75414	-1.32177	-0.75414

b. And through what we see above, we can derive the equation:

MULTIPLE LINEAR REGRESSION – CEREAL RATING

$$\begin{aligned} \text{Rating} = & 57.795 - 0.192 (\text{calories}) + 2.721 (\text{protein}) \\ & - 2.225 (\text{fat}) - 0.057 (\text{sodium}) + 2.477 (\text{fiber}) + \\ & 0.804 (\text{carbo}) - 1.038 (\text{sugars}) \end{aligned}$$

<i>calories</i>	energy (kcal)	<i>protein</i>	grams of protein
<i>fat</i>	grams of fat	<i>sodium</i>	mg of sodium
<i>fiber</i>	grams of fiber	<i>carbo</i>	complex carbohydrates
<i>sugars</i>	grams of sugars		

Or in other words, every cereal begins with a base score of 57.8.

Good

- Each gram of protein leads to a +2.72 increase in rating
- Each gram of fiber leads to a +2.48 increase in rating
- Each gram of complex carbs leads to a 0.8 increase in rating

Bad

- Each gram of fat is a -2.23 decrease in rating
- Each gram of sugar is a -1.04 decrease in rating
- Each calorie is -0.19 decrease in rating
- Finally, each milligram of sodium is a -0.06 decrease in rating

c. The adjusted R-squared is 0.9901. Meaning 99% of the variation of the model is explained. On top of this, each one of these predictors is statistically significant, passing the T-test ($P < 0.05$).

d.

Error Metric	Value
Sum of Squares of Error (SSE)	649.0051
Root Mean Square Error (RMSE)	2.9619
Mean Absolute Error (MAE)	2.2882
Standard Error of the Estimate	3.0456

Conclusion

This project set out to answer one question: What nutritional factors most accurately predict a high consumer rating for breakfast cereals? The answer is now clear. Healthier cereals win!!

From the descriptive analysis, patterns were obvious. Nabisco's wheat and bran-focused cereals topped the ratings, while General Mills' sugar-heavy lineup sat at the bottom. Lower-calorie and lower-sugar cereals consistently outperformed others, with shelf placement even reflecting this, where they were clustered on Shelf 3 alongside the highest ratings.

The regression model confirmed what the exploratory analysis suggested. With an adjusted R-squared of 0.9901, the model explains 99% of the variation in consumer ratings; an exceptionally strong fit. Every predictor was statistically significant, and the equation tells a

straightforward story: protein and fiber drive ratings up, while fat, sugar, calories, and sodium pull them down.

So, the takeaway is simple. If you want a better-rated cereal, build a healthier one. One that is high in protein and fiber but low in fat and sugar.

Video Description: <https://youtu.be/8ByWSSYA7Mw>

Constraints

- Small sample size of only 77 observations. This limits statistical power
- Pre-provided course dataset. I have no control over data collection methodology
- No demographic breakdown between ratings
- No test check for multicollinearity
- Very simplistic. The only tool used was XL miner, and only multiple linear regression was used to pull insights.